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$2NaOH + D_2O \rightarrow 2NaOD + HOD$

$HCl + D_2O \rightarrow DCl + HOD$

$NH_4Cl + 4D_2O \rightarrow ND_4Cl + 4HOD$

These exchange reactions are useful in determining the number of ionic hydrogens present in a given compound.

For example, when D_2O is treated with of hypo-phosphorus acid only one hydrogen atom is exchanged with deuterium. It indicates that, it is a monobasic acid.

$H_3PO_2 + D_2O \rightarrow H_2DPO_2 + HDO$

It is also used to prepare some deuterium compounds:

$Al_4C_3 + 12D_2O \rightarrow 4Al(OD)_3 + 3CD_4$

$CaC_2 + 2D_2O \rightarrow Ca(OD)_2 + C_2D_2$

$Mg_3N_2 + 6D_2O \rightarrow 3Mg(OD)_2 + 2ND_3$

$Ca_3P_2 + 6D_2O \rightarrow 3Ca(OD)_2 + 2PD_3$

4.6.2 Uses of heavy water:

1. Heavy water is widely used as moderator in nuclear reactors as it can lower the energies of fast neutrons
2. It is commonly used as a tracer to study organic reaction mechanisms and mechanism of metabolic reactions
3. It is also used as a coolant in nuclear reactors as it absorbs the heat generated.

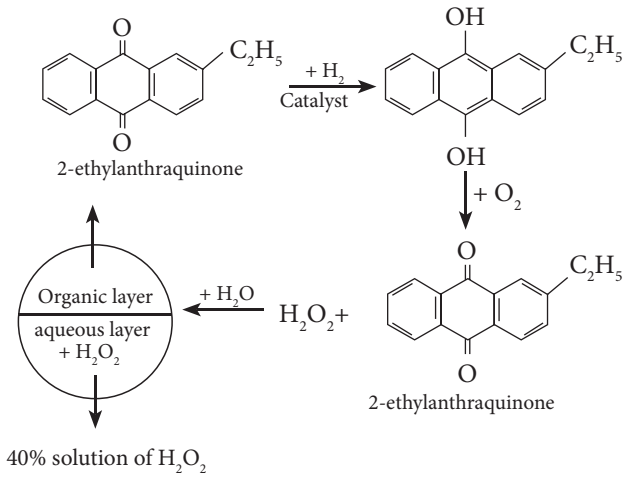
4.7 Hydrogen Peroxide:

Hydrogen peroxide (H_2O_2) is one of the most important peroxides. It can be prepared by treating metal peroxide with dilute acid.

$BaO_2 + H_2SO_4 \rightarrow BaSO_4 + H_2O_2$

$Na_2O_2 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O_2$

On an industrial scale, hydrogen peroxide is now prepared exclusively by autoxidation of 2-alkyl anthraquinol.



4.7.1 Physical properties:

Pure hydrogen peroxide is almost a colorless liquid (pale blue), less volatile and more viscous than water.

A 30 % solution of hydrogen peroxide is marketed as '100-volume' hydrogen peroxide indicating that at S.T.P., 100 ml of oxygen is liberated by 1 ml of this solution on heating.

4.7.2 Chemical properties:

Hydrogen peroxide is highly unstable and the aqueous solution spontaneously